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Title: Development and evaluation of supervised machine learning algorithms for early diagnosis of cardiovascular disease using predictive modelling

CARDIOCARE
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Introduction

Cardiovascular diseases (CVDs) represent a significant global health challenge, accounting for a substantial portion of morbidity and mortality worldwide. Timely identification of individuals at risk of developing CVD is essential for effective prevention and intervention strategies. Multiple risk factors should be properly incorporated, and more correlated nuances between the risk factors and outcomes should be determined. Therefore, the current management of CVD mainly focuses on preventive measures. Recent studies suggest that ~80% of premature CVD mortality could be prevented through early intervention. In addition, CVD has a slow onset and long incubation period; thus, it is generally at a more serious stage at the time of diagnosis. Therefore, early identification of high-risk groups for CVD is particularly important for its prevention and control.

Detecting CVDs prematurely is of great interest for the Healthcare Industry. According to the World Health Organization (WHO), almost 18 million deaths, representing 32% of global deaths in 2019 were caused by heart diseases, as part of CVDs. Furthermore, 75% of these cases took place in low or middle income countries, not to mention that 38% of premature deaths (under the age of 70) were caused by CVDs.

Cardiovascular disease is an umbrella term that includes related diseases such as coronary artery diseases or atherosclerosis, arrhythmias and hypertension. The primary cause of the disease occurs when the coronary arteries, which supply blood to the heart muscle, become narrowed or blocked by a buildup of plaque (atherosclerosis). This can lead to hypertension where the force of blood against the walls of the arteries is consistently too high. Over time, this can damage the arteries and increase the risk of heart disease, stroke, and other cardiovascular complications, Heart Failure where the heart cannot pump blood effectively, causing symptoms such as fatigue, shortness of breath, and fluid retention. Arrhythmias which are irregular heart rhythms. They can take the form of tachycardia (rapid heartbeat), bradycardia (slow heartbeat), or atrial fibrillation (an irregular and often rapid heartbeat).

Cardiovascular diseases often share common risk factors, such as a sedentary lifestyle, poor dietary habits, smoking, excessive alcohol consumption, and genetics. Identifying and managing these risk factors is crucial for preventing and treating CVD. Regular medical check-ups, lifestyle modifications, and, in some cases, medications or surgical interventions are key components of managing and preventing cardiovascular diseases.

Even though several approaches on how to detect these kinds of diseases using Machine Learning (ML) have been developed throughout recent times, in this work we aim to find a robust yet interpretable way of finding out which risk factors are more strongly correlated with CVDs, particularly coronary heart disease.

Problem Statement:

The problem is the high incidence of cardiovascular diseases (CVDs), such as heart attacks, strokes, and coronary artery disease, which continue to be a leading cause of morbidity and mortality worldwide. These diseases often strike without warning, and timely intervention is crucial for reducing the risk and severity. There is a need for a robust predictive model to accurately identify individuals at risk of CVDs, enabling early intervention and personalized risk assessment based on a diverse set of clinical, lifestyle, demographic, and genetic factors.

Aim

Obtaining results for the classification of heart diseases that allow us to have a high percentage of successes in the early detection of the disease. Our predictive model must achieve a high level of accuracy to provide meaningful risk assessments. Here we aim not only to find an accurate predictor but to examine which risk factors are more closely related to CVD detection, leading to prevention or better and early management. To pursue our goal, we use a public dataset from kaggle to test four (4) different machine learning models. We defined the importance of features in order to select the most relevant ones. Through a benchmark among these models, we determined the best model. Lastly, we implemented the best model in an easy application that can be used in preventing the detection of CVDs.

Objectives:

1. Identify patterns in the data that could help identify which factors have the biggest influence on a person's risk for developing heart disease. This could lead to better preventive health care measures for heart disease in general or for specific groups of people at higher risk than others (e.g. due to certain age or gender)
2. Develop a predictive model capable of accurately assessing an individual's risk of cardiovascular disease based on a comprehensive dataset of clinical, genetic, and lifestyle factors. Ensure the model's accuracy, sensitivity, and specificity to minimize both false positives and false negatives.
3. Create a user-friendly interface that can be integrated into healthcare systems for easy adoption by medical professionals.
4. Evaluate the model's performance and effectiveness in real-world clinical settings to demonstrate its potential to reduce CVD incidence and improve patient outcomes.

Key Challenges:

Data Complexity: Cardiovascular diseases are multifactorial, with risk influenced by a wide range of variables, including genetics, medical history, lifestyle, and demographic factors. Integrating and analyzing this diverse dataset can be challenging.

Data Quality: Ensuring the accuracy and reliability of data, which may come from various sources such as electronic health records, genetic testing, and wearable devices, is essential for the model's credibility.

Model Accuracy: The predictive model must achieve a high level of accuracy to provide meaningful risk assessments. Inaccurate predictions could lead to either unnecessary interventions or missed opportunities for prevention.

Bias Mitigation: To ensure fairness, the model should be developed with attention to potential biases related to demographics, race, or socioeconomic factors.

Addressing these challenges and objectives will lead to the development of a robust Cardiovascular Disease Prediction Model that can transform the approach to CVD prevention and management, ultimately saving lives and reducing the societal and economic impact of these diseases.

Value Proposition:

To address the challenges mentioned, a solution leveraging predictive modelling for early CVD risk identification can be highly beneficial. The value proposition is as follows:

1. Improved Patient Outcomes: By harnessing predictive modelling and data analytics, healthcare systems can identify individuals at risk of CVD at a much earlier stage. This early detection allows for timely interventions,

lifestyle modifications, and targeted treatment plans, ultimately leading to better patient outcomes and reduced morbidity and mortality.

2. Cost Reduction: Early diagnosis and intervention are more cost-effective than treating advanced-stage CVD. Identifying high-risk individuals early on enables cost-effective preventive measures, reducing the financial strain on healthcare systems and patients alike.

3. Enhanced Risk Assessment: Predictive modeling can incorporate a wide range of patient data, including genetic, lifestyle, and clinical factors, to provide a more accurate and comprehensive assessment of an individual's CVD risk. This improves the precision of risk prediction, especially in diverse populations.

4. Streamlined Healthcare: By integrating predictive modeling into healthcare systems, providers can create a more streamlined and coordinated approach to patient care. This facilitates better communication and sharing of patient information among primary care providers, specialists, and diagnostic facilities, ensuring timely CVD risk assessment and management.

In conclusion, leveraging predictive modeling for early CVD risk identification offers a powerful solution to bridge the existing gaps in the healthcare system, enabling early diagnosis, cost reduction, and improved patient outcomes. This approach not only benefits individual patients but also contributes to the overall efficiency and effectiveness of healthcare systems in addressing the growing global CVD epidemic.

Rationale

Cardiovascular Disease (CVD) is a leading cause of death and a significant burden on healthcare systems worldwide. As previously stated, CVDs encompasses various conditions, such as heart disease and stroke, and is often preventable with early detection and intervention. However, there are several pressing issues related to early CVD diagnosis and management:

- Late Diagnosis: In many cases, CVD is diagnosed at an advanced stage when the disease has already caused significant damage to the cardiovascular system. This late diagnosis can result in poorer patient outcomes, increased morbidity, and mortality rates.
- Fragmented Healthcare: Healthcare systems often operate in a fragmented manner, with limited communication and coordination between primary care providers, specialists, and diagnostic facilities. This can lead to delays in diagnosis and treatment, hindering the early identification of CVD risk factors.
- High Costs: Treating advanced-stage CVD is costly, placing a substantial economic burden on healthcare systems and patients. This financial strain is often preventable with early detection and lifestyle modifications.
- Limited Risk Assessment: Current risk assessment tools for CVD, such as the Framingham Risk Score, have limitations in accurately identifying individuals at risk, especially in diverse populations. These tools may not account for all relevant risk factors or evolving patient data.

Innovation

Wearable Devices: we aim to develop and use wearable sensors and smartwatches to continuously monitor physiological parameters, including heart rate, ECG, blood pressure, and activity levels. These devices can provide real-time data for risk assessment and early warning. This will in turn implement remote monitoring solutions that enable healthcare providers to track and manage patients' health remotely. Telemedicine and telehealth platforms can play a crucial role in this regard.

Innovations in cardiovascular disease prediction models have the potential to revolutionize prevention, early detection, and treatment, leading to improved patient outcomes and a reduction in the global burden of cardiovascular disease. These models can assist healthcare professionals in delivering personalized care, making predictions more accurate, and ultimately saving lives.

Impact of cardiovascular monitoring devices

1. Early Detection : They can detect abnormal heart rhythms or symptoms early, allowing for prompt medical intervention.
2. Data Tracking : Continuous monitoring provides a wealth of data about the patient's heart health over time, helping healthcare providers identify trends and make informed treatment decisions.
3. Remote Monitoring : Patients can be monitored from the comfort of their homes, reducing the need for frequent clinic visits and hospitalizations.
4. Treatment Adjustment: Healthcare providers can adjust medications or treatment plans based on real-time data, optimizing care for the patient's specific needs.
5. Improved Quality of Life : Patients have peace of mind knowing that their heart condition is being actively monitored, which can reduce anxiety and improve their overall quality of life.
6. Preventive Measures : Monitoring can help identify risk factors and lifestyle factors that may contribute to heart disease, enabling patients to make proactive changes to their habits and reduce the risk.

Business Model

Creating a successful business model for wearable health monitoring devices involves a strong focus on technological innovation, user experience, and partnerships within the healthcare ecosystem. Additionally, staying adaptable to changing market demands and regulations is essential for long-term success.

<table>
<tr><th colspan="5">Business Model Canvas</th></tr>
<tr>
<td rowspan="2">Key Partners

- Healthcare Institutions
- Data Providers
- Research Institutions</td>
<td>Key Activities

-Data Collection and Analysis
- Model Development and Training
- Customer Support and Training
- Marketing and Sales</td>
<td rowspan="2">Value Propositions

- Early Detection of Cardiovascular Diseases
- Improved Patient Outcomes
- Risk Mitigation for Insurers
- Research Support for Pharmaceutical Companies.</td>
<td>Customer Relationships

-Personalized Support
- Feedback Loop
-Continuous Updates</td>
<td rowspan="2">Customer Segments

-Patients with Cardiovascular
-Risk Healthcare Providers
- Insurance Companies Pharmaceutical Companies</td>
</tr>
<tr>
<td>Key Resources

- Machine Learning Algorithms
- Data Sources
- IT Infrastructure
- Domain Experts</td>
<td>Channels

- Direct Sales
- Online Platform Partnerships
- Marketing and Outreach</td>
</tr>
<tr>
<td colspan="2">Cost Structure

-Data Acquisition and Management
- Model Development
- Marketing and Sales
- Employee Salaries</td>
<td colspan="3">Revenue Streams

- Subscription Model
- Licensing
- Data Sales</td>
</tr>
</table>

Figure 1. Business model

Dataset

The attributes are Age, Sex, Chest Pain Type, BP (blood pressure), Cholesterol level, FBS over 120 (fasting blood sugar), EKG Results (electrocardiogram results), Max HR (maximum heart rate), Exercise Angina status, Slope of ST(slope of the ST segment on the ECG), Number of Vessels Fluroscopy (number of vessels seen on fluoroscopy) and Thallium Stress test results.

Understanding how each variable relates to heart disease risk will help you make better predictions based on this data set. For example, age is one variable that affects the odds of someone having a heart attack or stroke as it relates to arterial blockages- so it is important to note if age could be an independent factor or if other factors could be enhancing the odds in an individual patient's case . It should also be noted that although cholesterol levels are included in this data set , other laboratory parameters such as hdl cholesterol , triglycerides and ldl need to also be considered when assessing overall cardiovascular risk . Knowing gender can also play an important role when analysing possible trends for diagnosing new cases with suspected cardiac symptoms . Finally , understanding exercise angina status can provide critical information about a patient's history with exercise -induced chest pain which could result from myocardial ischemia

Column name	Description
Age	The age of the patient. (Numeric)
Sex	The gender of the patient. (Categorical)
Chest pain type	The type of chest pain experienced by the patient. (Categorical)
BP	The blood pressure level of the patient. (Numeric)
Cholesterol	The cholesterol level of the patient. (Numeric)
FBS over 120	The fasting blood sugar test results over 120 mg/dl. (Numeric)
EKG results	The electrocardiogram results of the patient. (Categorical)
Max HR	The maximum heart rate levels achieved during exercise testing. (Numeric)
Exercise angina	The angina experienced during exercise testing. (Categorical)
ST depression	The ST depression on an Electrocardiogram. (Numeric)
Slope of ST	The slope of ST segment electrocardiogram readings. (Categorical)
Number of vessels fluro	The amount vessels seen in Fluoroscopy images. (Numeric)
Thallium	The Thallium Stress test findings. (Categorical)
Heart Disease	Whether or not the patient has been diagnosed with Heart Disease. (Categorical)

Data processing

The data was cleaned, analysed, and pre-processed to ensure its quality and consistency. This involved handling missing values, dealing with outliers, and standardizing or normalizing the data as necessary.

There were some missing values in the database, and direct deletion of missing values resulted in the loss of sample information. Since there were a few variables with missing values in this study, continuous variables were filled using the mean, while categorical variables were filled using the mode. By standardizing continuous variables, categorical variables were processed by one-hot encoding to reduce the influence of different variable units and quantity levels on the analysis.

To examine the dataset, relationship esteem was determined between Target Diagnosis and each of the values. It tends to be noticed that the most correlated features with the target attributes were exercise-Induced Angina, and Chest Pain Type & ST Depression as shown in the heat map (figure). The heat map displaying the correlation between all characteristics is displayed to provide a clear picture of the feature connection between every one of the characteristics.

In addition, the heat map shows, that the highest correlated continuous feature which is Thallium Stress Test was plotted against age to see if there is any correlation. People with a higher thallium test findings

were more likely to get heart disease, regardless of their age. "Thallium Stress" which is a medical test used to evaluate the blood flow to the heart muscle. This test is often performed in the context of diagnosing or assessing cardiovascular diseases, particularly coronary artery disease. Thallium Stress Tests are typically used for the assessment of heart function: The test also provides information about how well the heart is pumping during rest and during stress (typically induced through exercise or medication). It can identify regions of the heart that may not function optimally due to reduced blood flow.

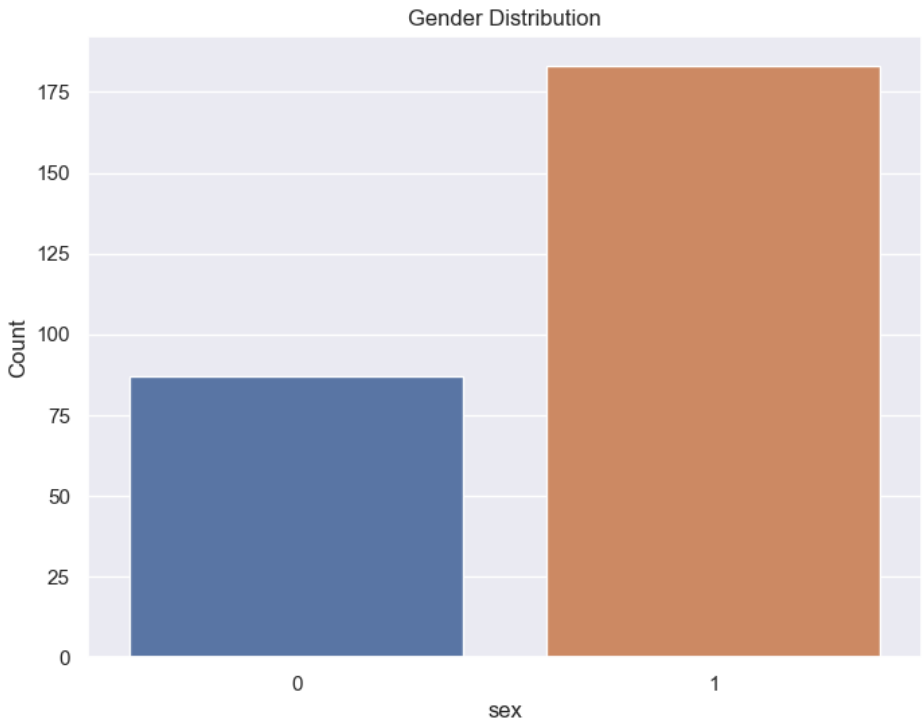


Figure 2. Gender distribution

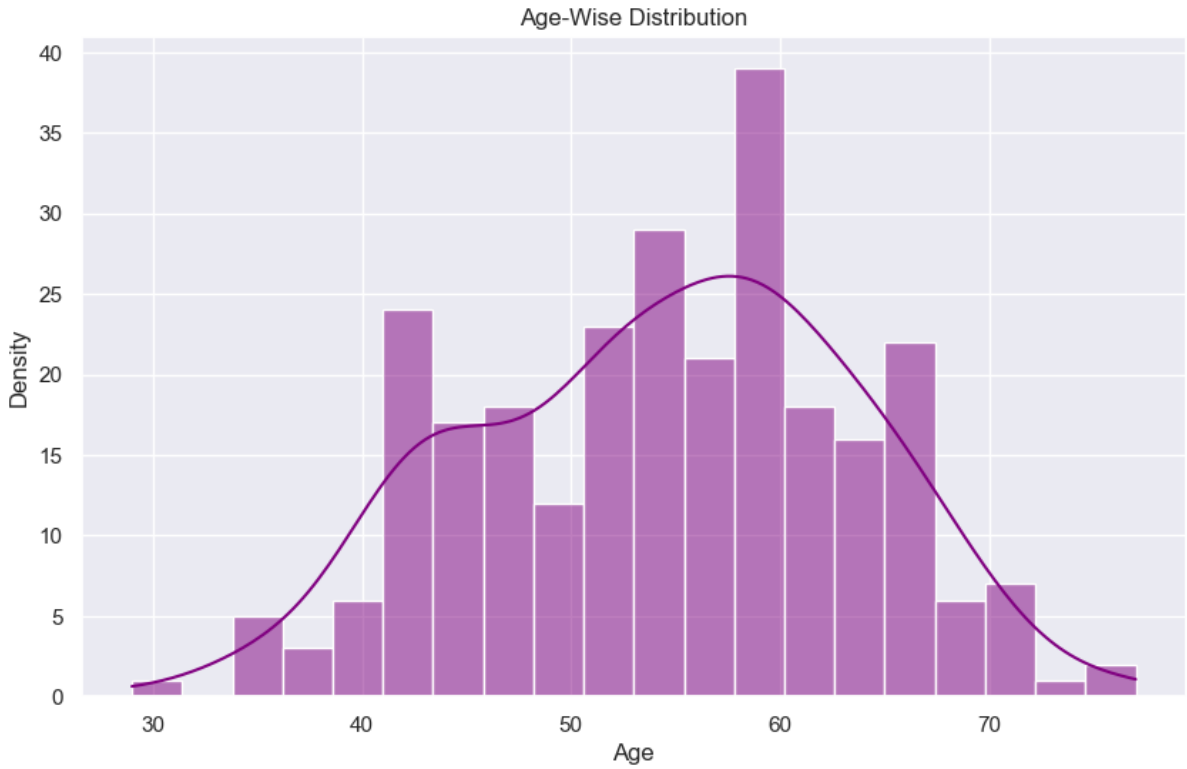
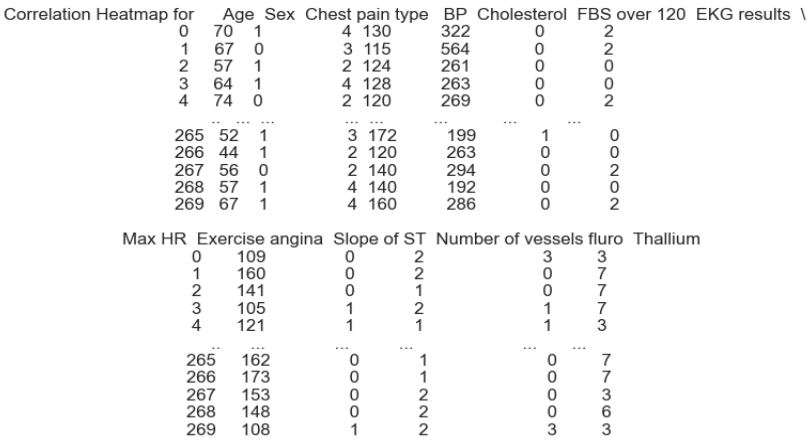


Figure 3. Age wise distribution



Correlation Heatmap for

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results \
0	70	1	4	130	322	0	2
1	67	0	3	115	564	0	2
2	57	1	2	124	261	0	0
3	64	1	4	128	263	0	0
4	74	0	2	120	269	0	2
..	...	...	...	...	...	...	...
265	52	1	3	172	199	1	0
266	44	1	2	120	263	0	0
267	56	0	2	140	294	0	2
268	57	1	4	140	192	0	0
269	67	1	4	160	286	0	2

	Max HR	Exercise angina	Slope of ST	Number of vessels fluro	Thallium
0	109	0	2	3	3
1	160	0	2	0	7
2	141	0	1	0	7
3	105	1	2	1	7
4	121	1	1	1	3
..	...	...	...	...	...
265	162	0	1	0	7
266	173	0	1	0	7
267	153	0	2	0	3
268	148	0	2	0	6
269	108	1	2	3	3

[270 rows x 12 columns]

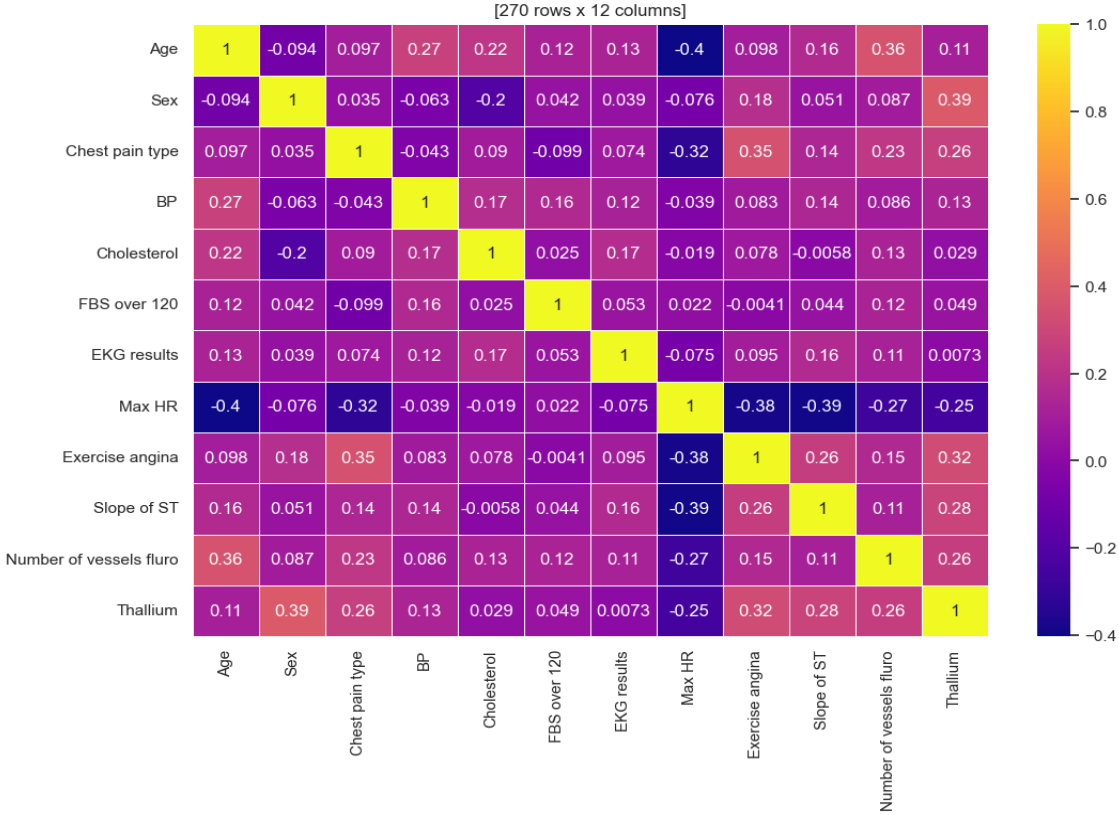


Figure 4. Correlation heat map

Proposed methodology: Machine learning

In the proposed ensemble model, firstly, the original data is fed into several different models. The Meta classifier is then utilized to estimate the input as well as the output of each model and also the weights are estimated. The top-performing models are chosen, while the rest are rejected. By utilizing a meta classifier, Stacking is the combination of numerous base classifiers learned on a single dataset using various learning methods. To make ensemble predictions, train the base learners and input the predictions into the meta learner, as shown in the flow diagram Figure 2. Figure 3 comprehensively shows our work i.e. first, we have taken Data from Kaggle which is shown in Data Collection & Pre-processing after deeply analyzing the data and finding correlations among other attributes we have removed outliers. Then splitted our data in 80% test and 20% train values.

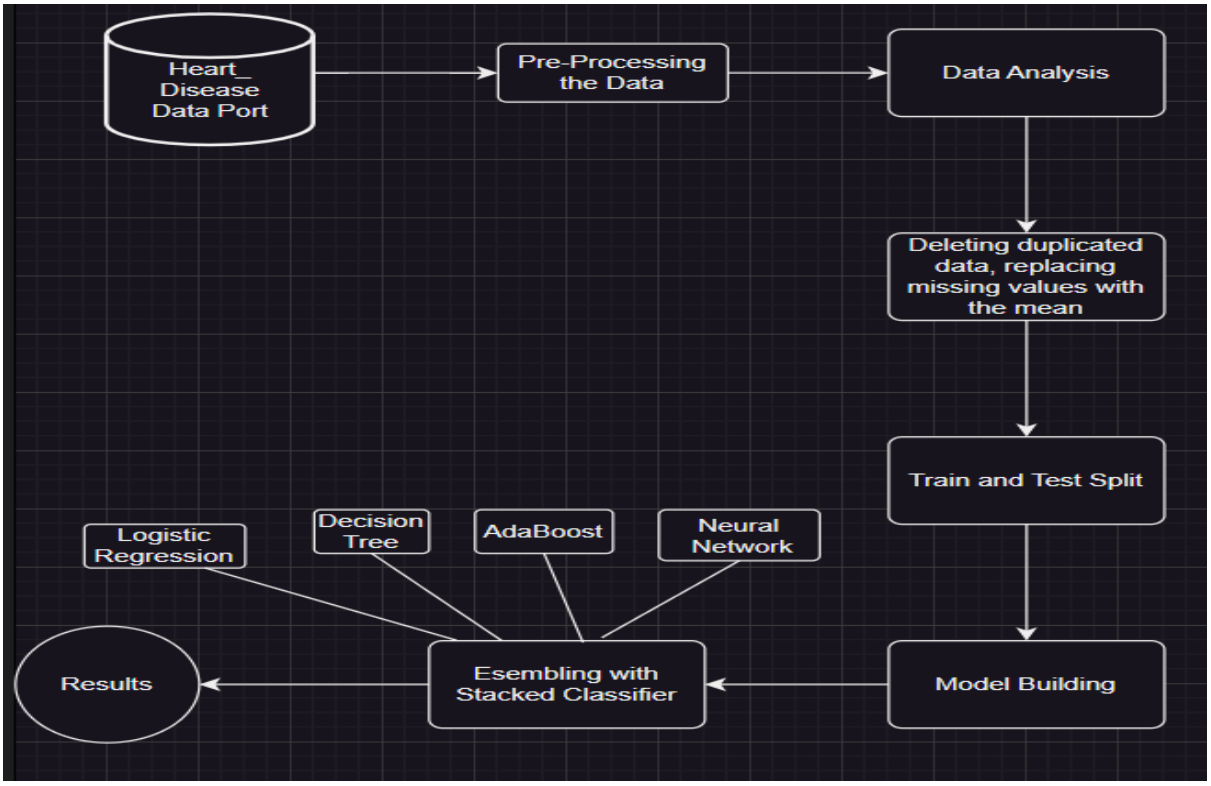


Figure 5: Methodology

Logistic regression

- Logistic Regression provides a straightforward and interpretable model. The coefficients of the logistic regression equation can be directly interpreted as the log-odds of the outcome variable. This makes it easier for healthcare professionals to understand and trust the model's predictions, which is crucial in a medical context.
- Logistic Regression can provide feature importance scores that indicate the strength and direction of the relationship between each feature and the disease risk.

AdaBoost

- AdaBoost's ensemble of weak learners can provide an effective tool for capturing complex relationships in clinical data, helping healthcare professionals identify patients at risk of cardiovascular disease based on a wide range of factors.

Decision tree

- Decision tree Gather a dataset with relevant features and a target variable indicating the presence or absence of cardiovascular disease.
- Perform cross-validation to ensure that the model's performance is robust and doesn't overfit the specific training/testing split.

Neural network

- Developing a cardiovascular disease prediction model using a neural network involves creating a deep learning model to handle complex, non-linear relationships within the data.
- Monitor the model's performance on the validation set during training to prevent overfitting. Consider using early stopping if validation performance plateaus.

Ensemble

- Ensemble methods are effective for capturing different aspects of the data and reducing overfitting. Creating an ensemble cardiovascular prediction model involves combining multiple machine learning models to improve predictive accuracy and robustness.

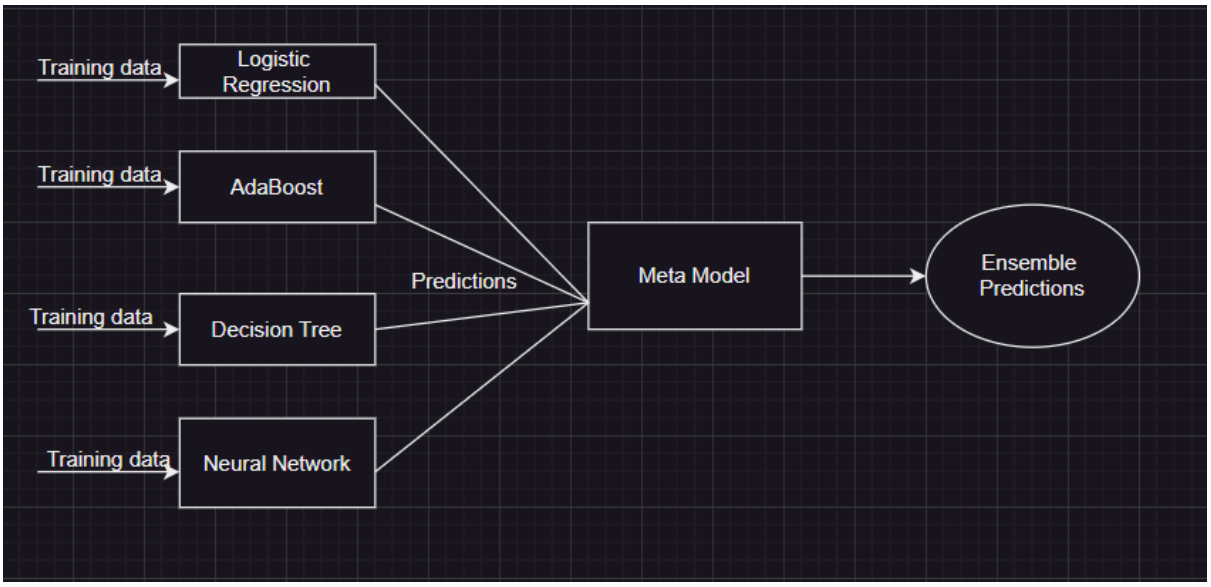


Figure 6. The proposed ensemble framework for heart disease prediction

Results and discussion

• The proposed model consisted of Stacking Based Ensemble learning which adds diversity to the classifier.

• A large dataset is used for training the ensemble model which helps in generalising the trained model.

• Hyper Parameter Tuning is used to select the best parameter for ML model training.

• The performance of the proposed models were compared on the the basis of accuracy, precision, sensitivity, precision, F1 Score and ROC.

All proposed models were used in the stacking approach. The accuracy of the baseline models is shown in Figure 2 and 3. As we have observed from figure 4 the best performing algorithms are Decision Tree, AdaBoost, Neural Network and lastly, Logistic regression. These algorithms are stacked as shown in Figure 3. As shown figure 4 and 5 , we have tested the algorithms based on accuracy, precision,sensitivity, specificity, F-1 Score to evaluate performance. From figure 5, the stacked classifier has a greater classification accuracy of 0.85 (85%) than the other classifiers. Considering other factors, the stacked classifier has the highest specificity score of 0.74 (74%) sensitivity score of (0.91) 91%, F1 Score of 0.77 (77%) and precision score of 0.91 (91%).

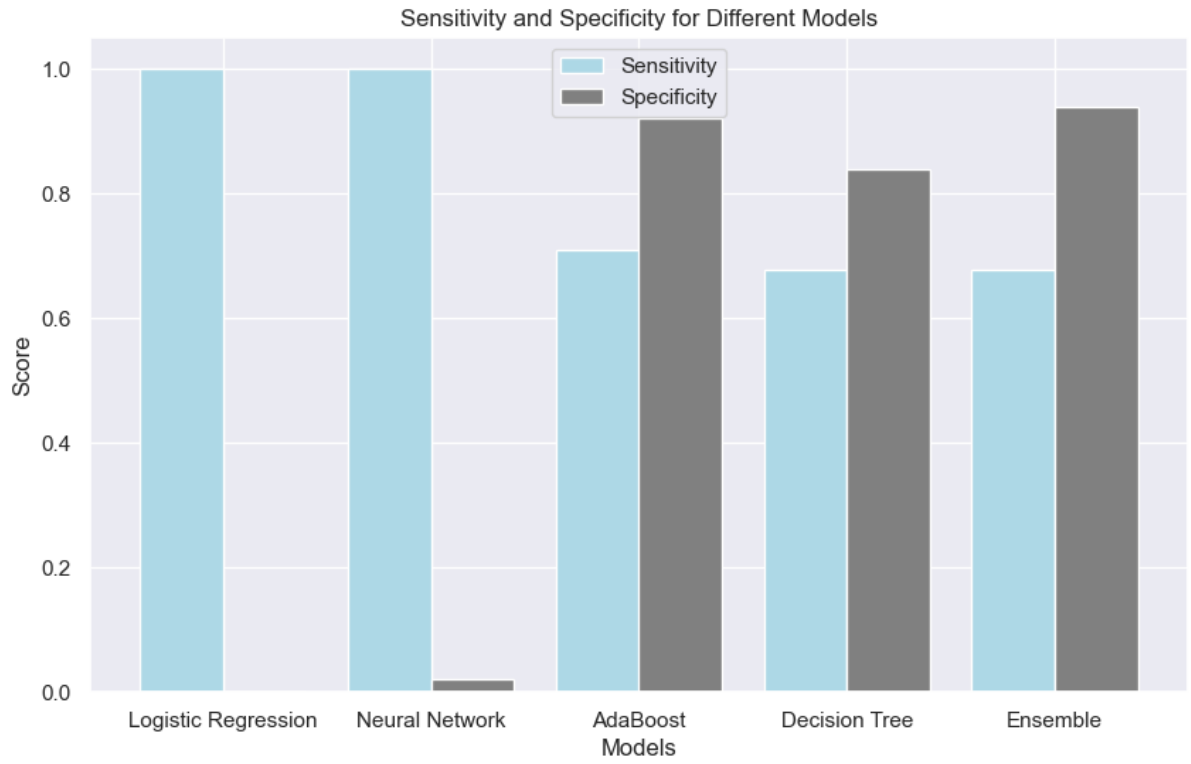


Figure 7: Model Comparisons

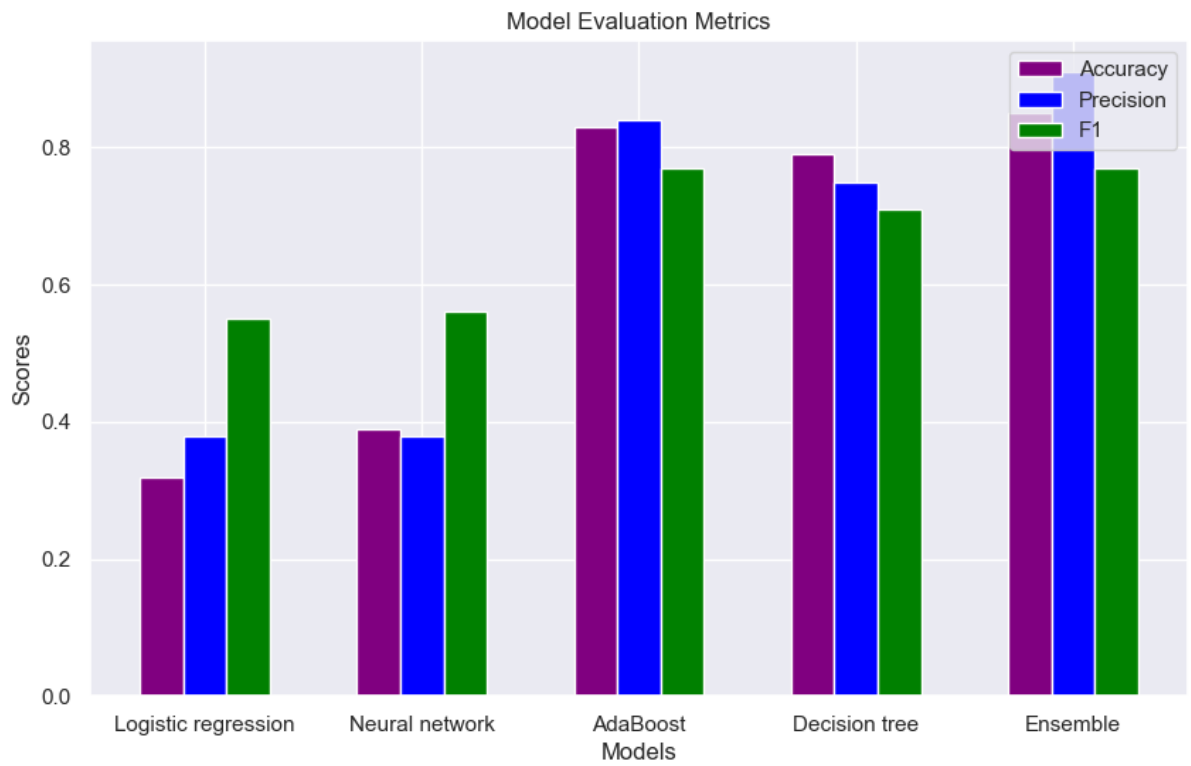


Figure 8. Model accuracy, precision and F1 scores.

Additionally, to analyse the models developed, a ROC (receiver operating characteristic curve) was constructed for each model, as shown in Figure 7. The ROC shows the classifier's diagnostic abilities, Figure 6 shows the area under each curve computed and presented. The better the model's diagnostic capacity, the

closer the area value of the ROC curve is to one. The precision-recall curve depicts the trade-off between precision and recall for different thresholds. A low false positive rate is associated with good accuracy, and a low false-negative rate is highly associated with recall. The comparison, in this case, shows an important part that can help medical research and diagnosis because of the relative nature of all the algorithms. Figure 6 completely shows the diverse nature and performances of all the algorithms. When we apply this stacked approach in real life, the medical community can get the benefit from this diversity by knowing how an algorithm performs in different cardiovascular diseases.

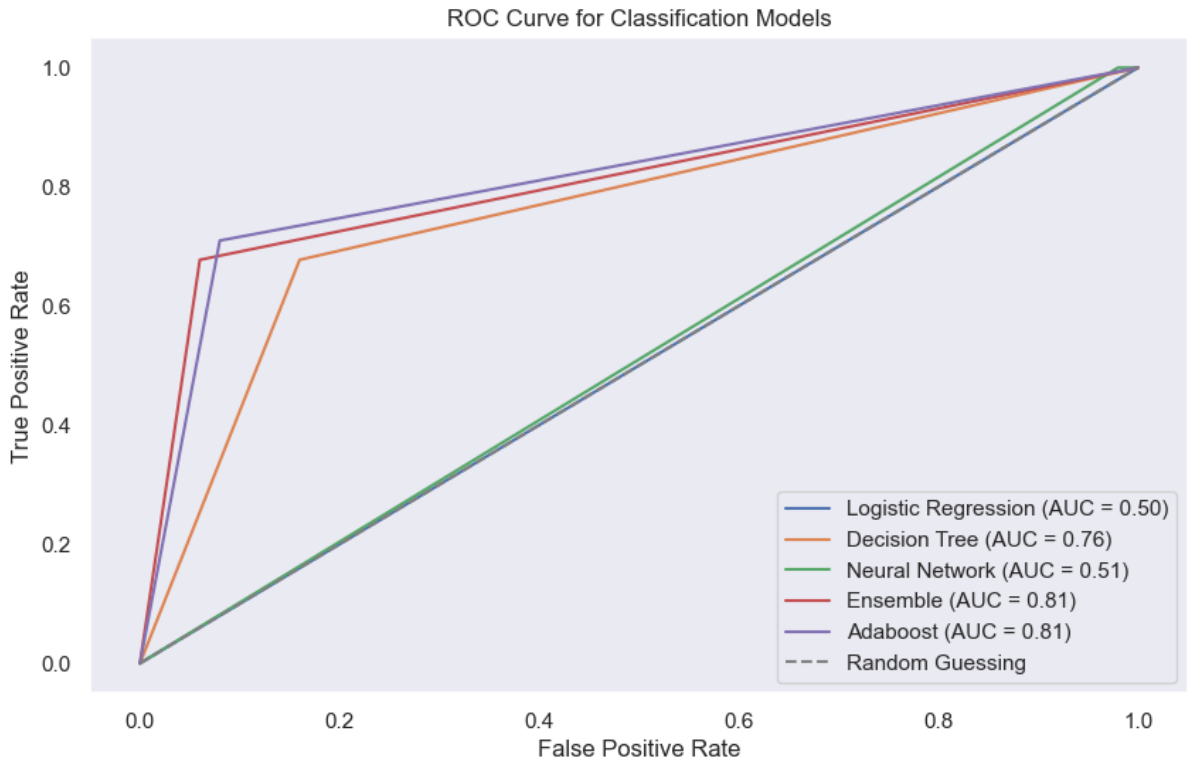


Figure 9. ROC analysis

Conclusion and future work

The development of a cardiovascular disease prediction model that correlates with the Thallium stress test and demonstrates high performance using an ensemble stacking machine learning approach is a significant achievement in the field of medical research and healthcare. This conclusion opens up several avenues for future work:

1. Validation and Clinical Trials: Before implementing the model in a clinical setting, it's crucial to conduct extensive validation and clinical trials. These trials should involve a diverse and representative patient population to ensure the model's real-world applicability.
2. Integration with Healthcare Systems: Integrate the predictive model with healthcare systems to provide real-time predictions and risk assessments for patients who have undergone the Thallium stress test. This integration should involve electronic health records (EHR) systems for seamless information sharing.
3. Continuous Monitoring and Wearable Devices: As you mentioned, the development of a wearable device for continuous cardiovascular health monitoring is a logical next step. This device can extend the model's capabilities by providing ongoing data, enabling early detection, and enhancing personalized treatment plans.

4. Long-Term Monitoring: Implement a system for long-term monitoring of patients, tracking changes in their cardiovascular health over time and adjusting treatment plans accordingly.